

SIMATS ENGINEERING

CO3 - ASSESSMENT TOOL 1

Case Study

COURSE NAME: DEEP LEARNING

COURSE CODE: MLA0408

COURSE OUTCOME COVERED

CO3: Analyze and apply convolutional neural network architectures, including convolutional layers, filters, parameter sharing, regularization, AlexNet and ResNet, to develop solutions for image classification and real-world computer vision applications. (L4)

CO Weightage: 25%

Assessment Tool Weightage: 25%

QUESTIONS:

Total Marks: 25

Case Study 1: CNN for Automated Medical Image Classification

A healthcare organization wants to develop a deep learning system to automatically classify chest X-ray images into **normal** and **disease-affected** categories. The dataset contains thousands of X-ray images with variations in size, brightness and image quality. A CNN model is proposed because it can automatically learn important visual features such as edges, textures, and abnormal regions without manually designing features.

Question:

1. Design and explain a suitable **CNN architecture** for this medical image classification system.
2. Describe the **motivation for using CNNs** and explain the role of **convolutional layers, filters, pooling layers, and fully connected layers** in the proposed architecture.
3. Explain how **parameter sharing** reduces the number of trainable parameters compared with a fully connected network.
4. Discuss suitable **regularization techniques such as dropout, data augmentation and batch normalization** to reduce overfitting.
5. Finally, explain whether **AlexNet or ResNet** would be more appropriate for this application and justify your choice based on feature learning, network depth, computational requirements and classification performance.

Case Study 2: CNN-Based Autonomous Vehicle Object Recognition

An autonomous vehicle company is developing a vision system that must identify **cars, pedestrians, traffic signs, bicycles, and road obstacles** from images captured by cameras mounted on the vehicle. The system must recognize objects under different lighting conditions, viewing angles, distances and road environments. The development team is considering **AlexNet and ResNet** architectures for building the CNN-based recognition system.

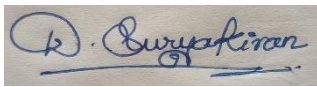
Question:

1. Analyze how a **CNN architecture** can be designed for this autonomous vehicle application.
2. Explain the purpose of the **input, convolution, activation, pooling, and fully connected/output layers** and describe how convolutional **filters progressively learn low-level and high-level features**.
3. Explain the importance of **parameter sharing** in reducing computational complexity and improving translation-related feature detection.
4. Discuss the possible problem of **overfitting** and recommend suitable regularization methods.
5. Compare **AlexNet and ResNet** in terms of architecture, depth, parameter efficiency, training difficulty, vanishing-gradient problems and feature-learning capability.
6. Finally, recommend the most suitable architecture for the autonomous vehicle system and justify your decision based on **accuracy, computational cost and real-time application requirements**.

Code of Conduct:

I D.Suryakiran /192325057(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.



Signature of the student:

Case Study 1: CNN for Automated Medical Image Classification

1. Suitable CNN Architecture

For chest X-ray classification into Normal and Disease-Affected, a CNN can be designed as follows:

Layer	Configuration	Purpose
Input	$224 \times 224 \times 1$	Grayscale X-ray image
Conv1	32 filters, 3×3 , ReLU	Detect edges and basic patterns
Batch Normalization	—	Stabilize and accelerate training
Max Pooling	2×2	Reduce spatial dimensions
Conv2	64 filters, 3×3 , ReLU	Learn textures and shapes
Batch Normalization	—	Improve training stability
Max Pooling	2×2	Down sample feature maps
Conv3	128 filters, 3×3 , ReLU	Learn complex abnormalities
Batch Normalization	—	Reduce internal variation
Max Pooling	2×2	Reduce computation
Conv4	256 filters, 3×3 , ReLU	Extract high-level features
Global Average Pooling	—	Convert feature maps to feature vector
Dropout	0.5	Reduce overfitting
Fully Connected	128 neurons, ReLU	Perform classification

Workflow:

X-ray → Preprocessing → Convolution → ReLU → Batch Normalization → Pooling → Convolution → Pooling → High-level Features → Fully Connected Layer → Sigmoid → Classification.

Before training, the images should be resized to a common size, normalized, and augmented using operations such as small rotations, translations and brightness changes.

Motivation and Role of CNN Components

Motivation for CNN

CNNs are suitable because chest X-rays contain important **spatial and visual patterns**. Instead of manually designing features, CNNs automatically learn hierarchical features:

- Early layers learn **edges and contours**.
- Middle layers learn **textures and anatomical structures**.
- Deeper layers learn **complex disease-related patterns**.
- The final layers use these features to classify the image.

Convolutional Layers

A convolutional layer applies small filters/kernels across the image.

For example, a 3×3 **filter** scans different regions and produces a feature map.

Different filters learn different patterns such as edges, textures and abnormal structures.

Filters

Filters are trainable matrices that detect specific visual features.

For example:

- Filter 1 $\rightarrow$ horizontal edges
- Filter 2 $\rightarrow$ vertical edges
- Filter 3 $\rightarrow$ textures
- Deeper filters $\rightarrow$ complex pathological patterns

The filters are learned automatically during training.

Pooling Layers

Pooling reduces the spatial size of feature maps.

Max pooling (2×2) selects the maximum value from each region.

Advantages:

- Reduces computational cost.
- Reduces the number of parameters in later layers.
- Provides some translation tolerance.
- Preserves important high-level features.

Fully Connected Layers

The extracted features are converted into a classification decision.

For binary classification, a **sigmoid output** can be used:

If the probability exceeds a selected threshold, the image can be classified as disease-affected; otherwise, it is classified as normal.

3. Parameter Sharing

Parameter sharing is one of the major advantages of CNNs.

Consider a 224×224 **grayscale image**:

If a fully connected layer with 1,000 neurons directly processes the image: weights are required, excluding biases.

In contrast, a 3×3 **convolutional layer with 32 filters** requires only: weights, excluding biases.

The same filter is reused across different locations of the image. This is called **parameter sharing**.

Therefore, CNNs:

- Require far fewer trainable parameters.
- Reduce memory requirements.
- Train more efficiently.
- Preserve spatial relationships.
- Reduce the risk of overfitting compared with a similarly sized fully connected network.

4. Regularization Techniques

Medical image datasets can still cause overfitting because the model may memorize training images instead of learning general disease characteristics.

a) Dropout

Dropout randomly disables a percentage of neurons during training.

For example:

means approximately 50% of selected neurons are temporarily ignored during each training step.

Benefit: Prevents excessive dependence on particular neurons and improves generalization.

b) Data Augmentation

New training examples can be generated by applying realistic transformations:

- Small rotations
- Horizontal/vertical translations
- Zooming
- Cropping
- Slight brightness/contrast changes
- Noise addition

This helps the CNN handle X-rays with different sizes, positions, brightness and image quality.

Important: Augmentation should be medically realistic. Excessive transformations could alter clinically important structures.

c) Batch Normalization

Batch normalization normalizes intermediate activations during training.

Benefits include:

- More stable training.
- Faster convergence.
- Better gradient flow.
- Some regularization effect.
- Reduced sensitivity to initialization.

Other useful techniques

Additional methods include:

- **Early stopping** – stops training when validation performance stops improving.
- **L2 regularization/weight decay** – discourages excessively large weights.
- **Class weighting** – useful when normal and disease classes are imbalanced.

For medical applications, performance should be evaluated using **sensitivity, specificity, precision, F1-score and ROC-AUC**, rather than accuracy alone.

5. AlexNet vs ResNet

Between AlexNet and ResNet, ResNet is more appropriate for this application.

Factor	AlexNet	ResNet
Network depth	Relatively shallow	Much deeper
Feature learning	Good basic features	More sophisticated hierarchical features
Architecture	Traditional CNN	Residual CNN

Factor	AlexNet	ResNet
Training deep networks	More difficult	Easier due to skip connections
Medical image feature extraction	Moderate	Generally stronger
Computational requirement	Lower	Higher
Classification performance	Good baseline	Usually better for complex visual patterns

Why ResNet?

Chest X-rays contain subtle abnormalities that may require learning **complex and hierarchical features**. ResNet uses **residual/skip connections**, which allow information and gradients to flow more easily through deep networks.

A simplified residual block is:

where:

- = input
- = learned transformation
- = shortcut connection

This makes it possible to train deeper networks effectively.

For example, **ResNet-50** can learn more complex representations than AlexNet while avoiding some of the optimization problems associated with very deep conventional CNNs.

Final Choice

ResNet is the preferred choice because:

1. It learns richer and more discriminative features.
2. Its greater depth is useful for subtle X-ray abnormalities.
3. Residual connections improve gradient flow.
4. It generally provides stronger image classification performance.
5. Transfer learning from a pretrained ResNet can reduce training time and data requirements.

However, **AlexNet may be preferred when computational resources are very limited** or when a simple baseline model is required.

Conclusion

A CNN-based system can effectively classify chest X-rays by automatically learning features from simple edges to complex disease patterns. **Convolution and parameter sharing** make the network computationally efficient, while **pooling** reduces spatial dimensions and **fully connected/output layers** perform classification. **Data augmentation, dropout and batch normalization** help control overfitting. For the proposed medical application, **ResNet is generally the better choice than AlexNet** because of its deeper feature-learning capability and residual connections, although it requires greater computational resources.

Case Study 2: CNN-Based Autonomous Vehicle Object Recognition

1. CNN Architecture Design

For an autonomous vehicle, the CNN should recognize multiple objects such as cars, pedestrians, bicycles, traffic signs, and obstacles under different environmental conditions.

A suitable architecture can be:

Layer	Configuration	Purpose
Input	$224 \times 224 \times 3$ RGB image	Accept camera image
Conv1	32 filters, 3×3 , ReLU	Detect edges and colors
Max Pooling	2×2	Reduce spatial size
Conv2	64 filters, 3×3 , ReLU	Detect textures and simple shapes
Max Pooling	2×2	Reduce computation
Conv3	128 filters, 3×3 , ReLU	Learn object parts
Batch Normalization	—	Stabilize training
Conv4	256 filters, 3×3 , ReLU	Learn high-level features
Global Average Pooling	—	Reduce feature dimensions
Fully Connected	256 neurons	Combine learned features
Dropout	0.5	Reduce overfitting
Output	Softmax	Classify objects

For actual autonomous driving, **object detection** is more appropriate than simple image classification because the system needs to determine **what the object is and where it is**. Therefore, a CNN backbone such as ResNet can be combined with an object-detection framework.

Example:

Camera → Image Preprocessing → CNN/ResNet Backbone → Feature Maps → Detection Head → Bounding Boxes + Object Classes

Example output:

Object	Location	Confidence
Car	Bounding Box 1	98%
Pedestrian	Bounding Box 2	94%
Traffic Sign	Bounding Box 3	91%
Bicycle	Bounding Box 4	89%

2. Purpose of CNN Layers and Progressive Feature Learning

Input Layer

The input layer receives RGB camera images.

For example:

The images can be resized and normalized before being given to the CNN.

Convolutional Layer

Convolution applies trainable filters to different regions of the image.

A filter can detect:

- Edges
- Corners
- Lines
- Color patterns
- Textures

The convolution operation can be represented as:

where I is the input and F is the convolution filter.

Activation Layer

A ReLU activation function is commonly used:

It introduces non-linearity and allows the network to learn complex patterns.

Pooling Layer

Max pooling reduces the spatial dimensions of feature maps.

For example:

reduces the feature-map size while retaining strong features.

It helps:

- Reduce computation.
- Reduce memory requirements.
- Provide some translation tolerance.
- Retain important features.

Fully Connected/Output Layer

The fully connected layer combines the extracted features.

The final output layer produces object classes using **Softmax** for mutually exclusive classification:

For object detection, the output additionally includes **bounding-box coordinates and confidence scores**.

Progressive Feature Learning

CNN filters learn features hierarchically:

Early layers:

Edges → Lines → Corners → Colors

Middle layers:

Textures → Shapes → Wheels → Windows → Sign patterns

Deep layers:

Car → Pedestrian → Bicycle → Traffic Sign → Road Obstacle

Thus, the network automatically learns increasingly complex visual representations.

3. Importance of Parameter Sharing

A CNN uses the **same filter weights at different locations** of an image.

For example, a 3×3 filter does not need separate parameters for every position. The same filter can detect a vertical edge whether it appears on the left, center, or right side of the image.

For a single 3×3 filter with three RGB channels:

weights are required, plus one bias.

If 64 such filters are used:

weights are required.

A fully connected network would require vastly more parameters because every input pixel would connect to every neuron.

Advantages

Parameter sharing:

- Reduces the number of trainable parameters.
- Reduces memory usage.
- Reduces computational requirements.
- Makes training faster.
- Allows the same visual feature to be detected at different image locations.

This is especially important for autonomous vehicles because objects can appear **anywhere in the camera frame**.

4. Overfitting and Regularization

Autonomous vehicle datasets contain many images, but the model can still overfit to particular roads, weather conditions, camera positions, or locations.

For example, the model may perform well on sunny roads but poorly on rainy or nighttime roads.

Suitable techniques

1. Data Augmentation

Training images can be modified using:

- Rotation
- Scaling
- Cropping
- Translation
- Brightness changes
- Contrast changes
- Gaussian noise
- Weather simulation

This improves robustness to different driving conditions.

2. Dropout

Dropout randomly disables neurons during training.

For example:

This prevents the network from becoming overly dependent on particular neurons.

3. Batch Normalization

Batch normalization normalizes intermediate activations.

It provides:

- Stable training.

- Faster convergence.
- Better gradient flow.
- Some regularization.

4. L2 Regularization

L2 regularization penalizes excessively large weights:

This encourages simpler models.

5. Early Stopping

Training is stopped when validation performance stops improving.

6. Transfer Learning

A pretrained CNN can be fine-tuned using a vehicle-specific dataset. This reduces training requirements and can improve performance when labeled data is limited.

5. AlexNet vs ResNet

Feature	AlexNet	ResNet
Architecture	Traditional CNN	Residual CNN
Depth	8 learnable layers	Can have 18, 34, 50, 101, 152+ layers
Feature learning	Good	Excellent for complex features
Parameter efficiency	Less efficient	Better depth-to-performance trade-off
Training difficulty	Relatively easier	Easier to train deep networks because of skip connections
Vanishing gradients	More susceptible in deeper versions	Significantly reduced
Skip connections	No	Yes
Computational cost	Lower	Generally higher
Accuracy	Good baseline	Usually higher
Complex object recognition	Moderate	Strong
Transfer learning	Available	Very effective
Autonomous driving suitability	Basic systems	More suitable for advanced systems

Alex Net

Alex Net is a relatively shallow CNN consisting of convolutional and fully connected layers. It demonstrated the effectiveness of deep CNNs for image recognition.

Advantages:

- Simple architecture.
- Lower computational cost.
- Faster inference than many deeper networks.
- Suitable for basic classification.

Disadvantages:

- Less powerful feature representation.
- Large fully connected layers.
- Less suitable for complex visual environments.
- No residual/skip connections.

ResNet

ResNet introduces **residual/skip connections**.

A residual block can be represented as:

Instead of forcing the network to learn the entire transformation, it learns a residual function .

This helps information and gradients pass through deep networks and reduces the **vanishing-gradient problem**.

ResNet can therefore learn more complex features such as:

- Pedestrian shapes.
- Vehicle structures.
- Traffic-sign patterns.
- Road obstacles.
- Objects under different viewpoints.

6. Recommended Architecture

ResNet is the recommended architecture.

For autonomous vehicles, accuracy and robustness are extremely important because incorrect recognition of a pedestrian, vehicle, or obstacle can create a safety risk.

Reasons for selecting ResNet

1. Higher accuracy

ResNet's deeper architecture can learn complex and subtle visual features from road scenes.

2. Better feature learning

It progressively learns:

This is useful for distinguishing visually similar objects such as bicycles, motorcycles, pedestrians, and road signs.

3. Reduced vanishing-gradient problem

Skip connections allow gradients to propagate effectively through deeper networks.

4. Better generalization

With appropriate augmentation and regularization, ResNet can handle variations in:

- Lighting
- Weather
- Distance
- Viewpoint
- Road environment

5. Transfer learning

A pretrained ResNet can be fine-tuned on autonomous-driving datasets, reducing training time and data requirements.

Computational consideration

The main disadvantage is that a large ResNet can require more computational resources than AlexNet. Since autonomous vehicles require **real-time inference**, using a very large model may increase latency.

Therefore, a practical choice would be a moderately sized ResNet, such as ResNet-18 or **ResNet-34**, or an optimized ResNet variant, deployed on dedicated vehicle hardware.

Final Decision

Reason: ResNet provides stronger feature learning and higher recognition accuracy while its residual connections allow deep networks to be trained effectively. For real-time autonomous driving, a lightweight ResNet variant provides a better balance between accuracy, computational cost, and inference speed than using a very large ResNet or the older AlexNet architecture.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding ; some issues missed	Poor understanding; problem not identified correctly
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning
Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided
Clarity	10	Clear, well-structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand